



GRAVITY
ELEVATING SOLUTIONS

ABOUT US

Gravity was founded in South Africa in 1999. Since then, Gravity has been a leader in the work-at-height industry, assisting in the development of a variety of standards, providing excellent training, and producing innovative solutions for worksites and a range of fall-arrest equipment.

Gravity is committed to providing quality services and products, with the clients' satisfaction always being the highest priority.

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H. The Way Forward

General site safety

1. Always wear seat belts when travelling in vehicles.
2. Always use suitable personal protective equipment (PPE), a safety harness and fall protection equipment when working at height (Harnesses must be attached to the structure/anchor points at all times when working at height.)
3. Never carry out electrical work on electrical equipment unless competent to do so.
4. Never work under the influence of substances.
5. Never exceed the road speed limits.
6. Never use a handheld devices whilst driving.



7. Always wear a helmet when travelling on a motorbike.



8. No passengers are allowed on the back of pickup's or trucks.



9. Never undertake any street or underground work activities unless competent to do so.



10. When working in the proximity of power, always maintain the required safe distance and use the correct insulated equipment.



11. Never work without competent supervision, the supervisor must be able to assess, manage and control the risks on the work site.



12. Ensure medical fitness for all personnel on site as per job scope.



A. Fall Protection Plan

Before any work at height may commence, all the fall risks on site must be addressed and documented in a fall protection plan (FPP).

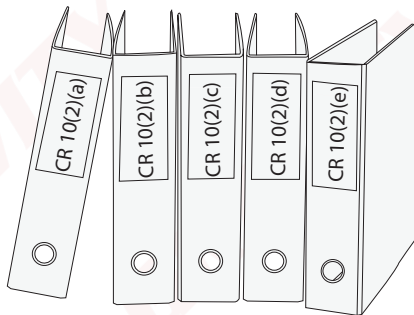
According to CR10(2) the Fall Protection Plan must include:

- CR10(2)(a) - **Risk Assessment**
- CR10(2)(b) - **Medical Fitness**
- CR10(2)(c) - **Training Programme**
- CR10(2)(d) - **Equipment Management**
- CR10(2)(e) - **Rescue Plan**

An FPP must also be reviewed after any incident occur, changes on site, or should the work scope change.

A FPP must be updated to ensure that all fall risks are mitigated and controlled on site.

Note: All this must be done BEFORE work may start.



1. Risk Assessment

The most important part of the Fall Protection Plan is the **Risk Assessment**.

What to do when approaching a work site

STEP 1 : RISK IDENTIFICATION

The following are factors that must be considered when identifying risks:

- **Site requirements**
- **Access method requirements**
- **Required team composition**
- **Fall protection requirements**
- **PPE requirements**
- **Environmental**

After this, the risks must be communicated to everyone on site by means of:

- Site induction
- Toolbox talk

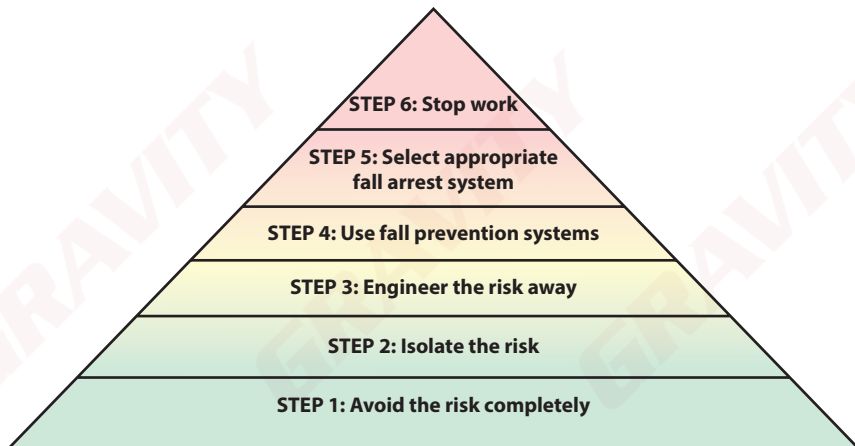
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STEP 2 : PRE-MITIGATION RISK RATING

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	LOW	LOW	LOW	LOW	LOW
	Unlikely	LOW	LOW	LOW	MEDIUM	MEDIUM
	Possible	LOW	LOW	MEDIUM	MEDIUM	HIGH
	Likely	LOW	MEDIUM	MEDIUM	HIGH	HIGH
	Very Likely	MEDIUM	MEDIUM	HIGH	HIGH	HIGH

No work is allowed to commence if there is a **Medium** or **High** risk present on site.
Thus, we have to mitigate all identified risks in order for them to receive a **Low** risk rating.

STEP 3 : RISK MITIGATION HIERARCHY



Hazardous Environmental Conditions:

No work-at-height shall be conducted during any inclement weather conditions. Should the situation change whilst at height, all technicians must safely descend to the ground and the Site Supervisor must report back immediately to the Project Manager.

- Heat
- Fumes

Examples of inclement weather conditions:

- Rain
- Lightning
- Extremely cold weather
- Extremely hot weather
- Sleet or snow
- High or gusty winds
- Poor visibility (fog or smog)



- **Additional Hazardous Environmental Conditions:**

Bee awareness

Honey Bees are an endangered species, but are also a hazard in the work-at-height industry.

They are responsible for pollinating 70 of the 100 crop species that feed 90 % of the world and this is why we should not kill them, but rather protect them.



The Do's and Don'ts of bees.

First, you need to know when Bees are most active "Prime Time" for the swarms are during spring and summer time.

When finding a bee colony unexpectedly:

- Do not approach closer.
- Do not make sudden movements.
- Slowly but efficiently move away.
- Bees have a way of warning you in different ways, by fly/bump into you or they will fly more actively/ making more noise.
- Listen and adhere to the warnings.
- Do not swat at bees, this will only cause them to be aggravated.
- Stay calm. Bees can feel if you are stressed and then they will get stressed.



If you are attacked by bees:

- Do not stand still and swat at them
- Run away in a straight line and do not zig-zang, you will cover a bigger area of ground running in a straight line.
- Keep running for at least a KM or to the closest enclosed space.
- If not in an urban area run into a dense brush, pull the thickest brush around your head. A few will sting but most bees will relent and move away.
- Do not scream and panic.
- Bees will enter your mouth and sting you.
- Try and keep your head covered.
- If you manage to take shelter with a locked door, switch off the lights. Bees are attracted to light.
- Do not jump into water. The bees will wait until you come up to breathe.



DO NOT:

- Throw things at bees.
- Try to burn them out.
- Try to smoke them out.
- Try to poison or kill the bees.
- Attempt to move the bees yourself.

**It is important to:**

- Always observe your surroundings.
- Remember that bees are everywhere.
- If you do get stung, remove the sting by scraping the surface of the sting with a Card or something similar.
- Do not pull out the stinger, by pulling/ pinching on it you may press the venom sack.
- Call a professional to move the bee colony.
- Remember that Smoking bees is a delicate process.
- If you are allergic ensure to have an “Epipen” to prevent you from going into Anaphylactic Shock.

**When Visiting a site:**

- Check for any quick movement of flying, this is characteristic of the honey bee.
- If observed stop and look for any activity, the activity will give an indication of where the colony made a nest.

After identifying the nest, the following needs to be done:

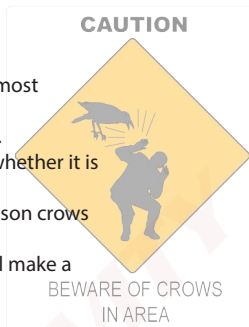
- Take photos of where the bees have made their nest.
- Inform your management about the nest in order for him to get a professional to move the bees

Additional Hazardous Environmental Conditions:

Bird awareness

Crows

- Crows are found on every continent except Antarctica, and are among the most intelligent animals on the planet.
- They are able to recognize a single person's face among many other people.
- They are able to even communicate to other crows and share information whether it is good or bad.
- Crows attack humans because they enter their habitat, during breeding season crows defend their eggs or young ones to humans that threaten them.
- If you see a nest or you see a crow on the structure do not advance they will make a noise to warn you before they attack.
- Contact a professional to come and take care of the animal.



Hawks and Eagles

- Hawks and Eagles are also among the most intelligent birds on the planet, but are also a danger people working at height.
- With most birds of prey you stand a chance of being attacked due to specific reasons.
- The most common reason is, it is breeding season and you are close to their nest posing a danger to their eggs or young ones.
- Hawks and Eagles will warn you by making noise before they attack.
- By entering their habitat/lifestyle you automatically pose a threat to them because they think you want to injure them and take their food and or water, so it is within their nature to protect themselves.
- Do not attempt to deal with them by yourself, get a professional to come to take care of the animal.
- Inform your manager about them and take photos as prove.



STEP 4 : POST-MITIGATION RISK RATING

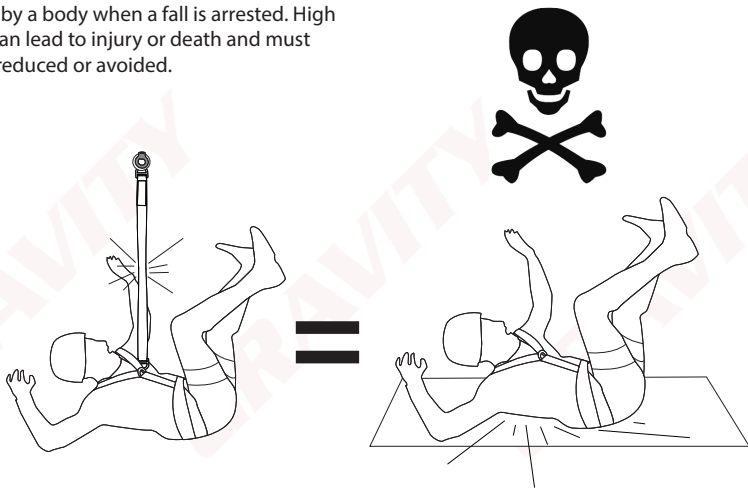
		HAZARD SEVERITY				
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B. Work at Height Basics

1. Shock Load

The force felt by a body when a fall is arrested. High shock loads can lead to injury or death and must therefore be reduced or avoided.



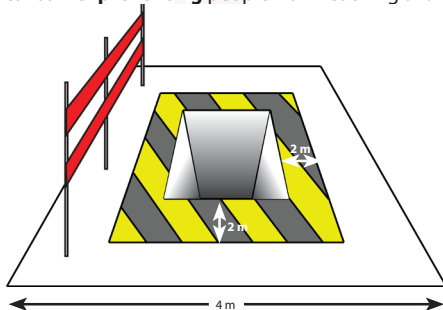
C. Site Establishment.

1. Before Work Starts

- Ensure that you comply with the client requirements on site. E.g. night work is allowed etc.
- Before work starts, ensure that there are at least 2 workers on site with the ability to rescue each other.
- Ensure that the fall protection system is relevant and fit for purpose as per site and task.
- Ensure that a comprehensive risk assessment has been done and that all workers have been inducted on the risk assessment.
- Ensure that all equipment is inspected before use and safe for use and fit for purpose.
- Ensure that a rescue plan is in place and a sufficient rescue kit is on site with a first aid kit.
- Ensure that the structure/site has been inspected and deemed safe for use
- Ensure that should traffic be a risk that a traffic management plan is in place.

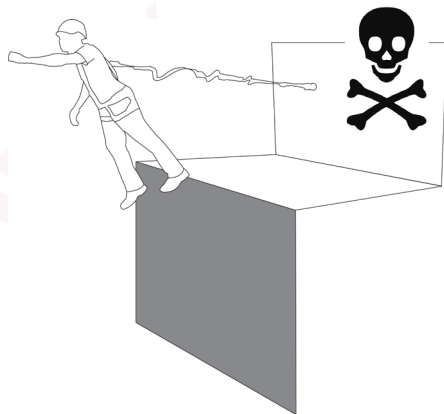
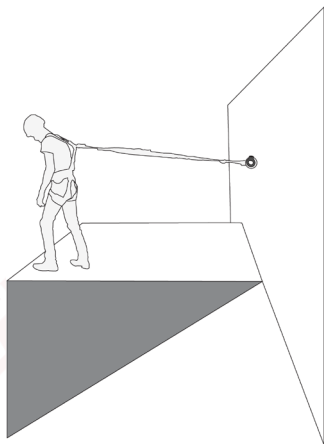
2. Exclusion Zones and Barricading

- **Exclusion zones:** A zone or space where people are **not allowed** to walk or work within, for example, a 2 m space around a hole in a roof.
- **Barricading:** A physical barrier **preventing** people from reaching a fall risk position.



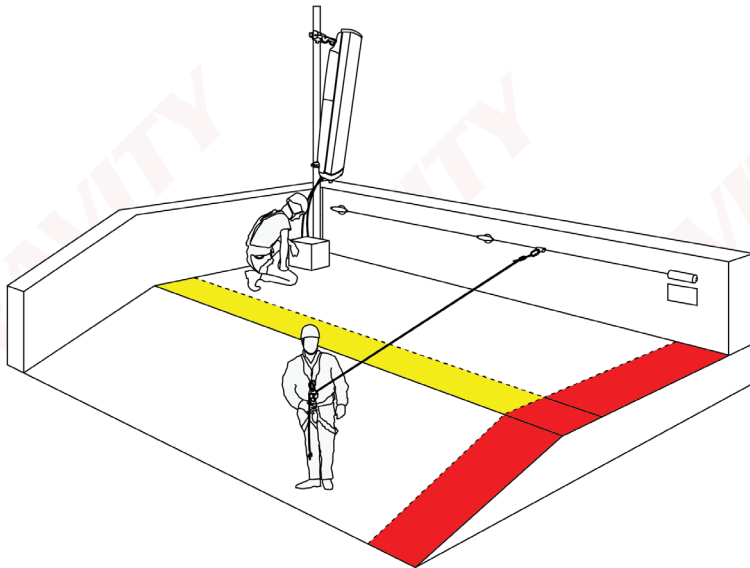
3. Work Restraint

A system where we connect a worker to the structure in such a way that it prevents him/her from reaching the edge, thus preventing a fall.



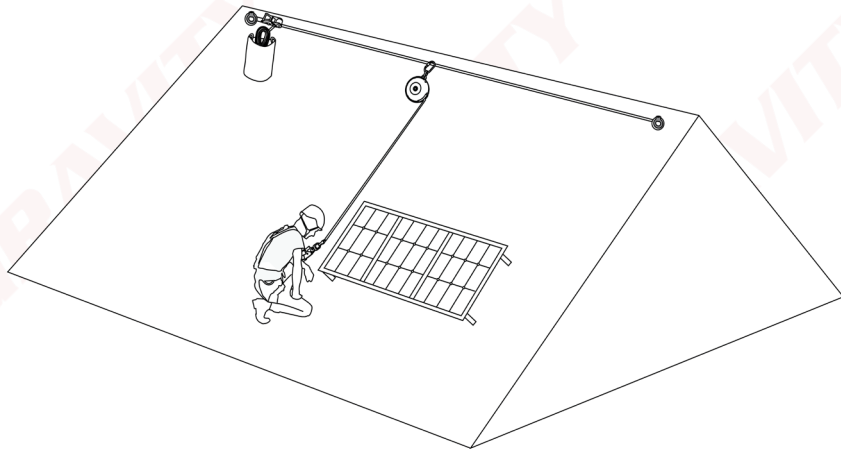
4. Roof Top Work Restraint System

One can combine the horizontal lifeline with a work restraint system to create a system ideal for rooftop work.



5. Pitched Roof Work

One can combine the temporary horizontal lifeline with a retractable lifeline to create a system ideal for pitched roof work.



6. Fall Arrest

A system used to suspend a person safely AFTER a fall has occurred, and doing all this in such a way as to prevent injury as far as possible.

Fall Arrest does NOT prevent a fall like Work Restraint. It only prevents a person from serious injury or hitting the ground due to a fall by reducing the shock load.



7. Full Range of Personal Protective Equipment (PPE)

In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment, namely:

- **Identification** - The name and purpose of each piece of equipment.
- **Certification** - Only equipment with the correct South African National Standards (SANS) or European National Standards (EN) should be used and must be purchased through a reputable retailer such as Gravity Training. For example, Gravity Training only makes use of ISO 9001:2015 certified equipment suppliers.
- **Limitation** - How to use and not use equipment, the safe use criteria and hazards to identify when using each piece of equipment, and the maximum load that each piece of equipment is certified to carry. Equipment load-bearing ability is given by the MANUFACTURER as **Safe Working Load** and the **Breaking Strength** and should never be exceeded.
- **Inspection** - Pre-use inspection should be done before the start of work and that equipment should be sent to the appointed competent equipment controller for detailed inspections at least every 6 months. The use of faulty or damaged equipment may lead to serious injury or death.
- **Note that the S.W.L will differ from one manufacturer to another.**

Climbing Helmet (EN 12492)



Construction Helmet (EN 397)



A helmet is used to protect the technician's head during a fall as well as from falling objects:

- Must have chinstrap.
 - Must be on the worker's head at all times.
 - Chinstrap is designed to break under a force of 50 kg.
 - Is designed in the a way that it can take top impacts and side impacts.
-
- Do not use an EN 397 Helmet for working at height .
 - Reasons why one should not use these helmets, they are not rated for side or back impact forces.
 - Should an object fall on the front of the helmet, it will lift up the back of the helmet and expose the user's head to potential injury.

Full Body Harness (EN361, 358)



A harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- Fall arrest systems must preferably be connected the chest D-link.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.
- S.W.L. = 140 kg
- M.B.S. = 1400 kg

Donning and duffing of a harness

When donning a harness ensure the following:

The waistbelt is above your hips and tight.

The shoulder straps are tight so that you can fit a fist between your chest and the harness.

The leg loops are adjusted so that they do not restrict blood flow when you crouch.

All the buckles are connected and secure.

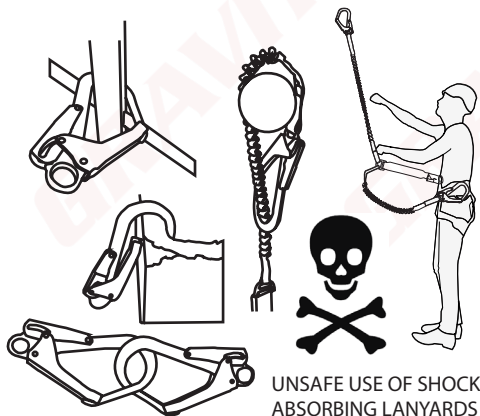


Shock Absorbing Lanyard (EN 355)

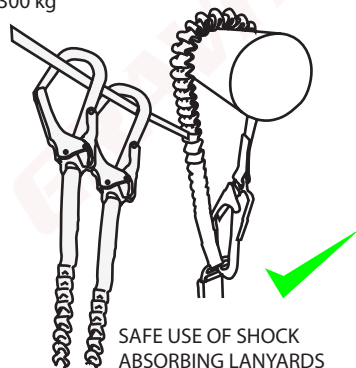


A shock absorber acts as the Primary safety when working at height.

- Shock absorber rips open (deploys) to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.
- S.W.L. = 130 kg
- M.B.S. = 1300 kg

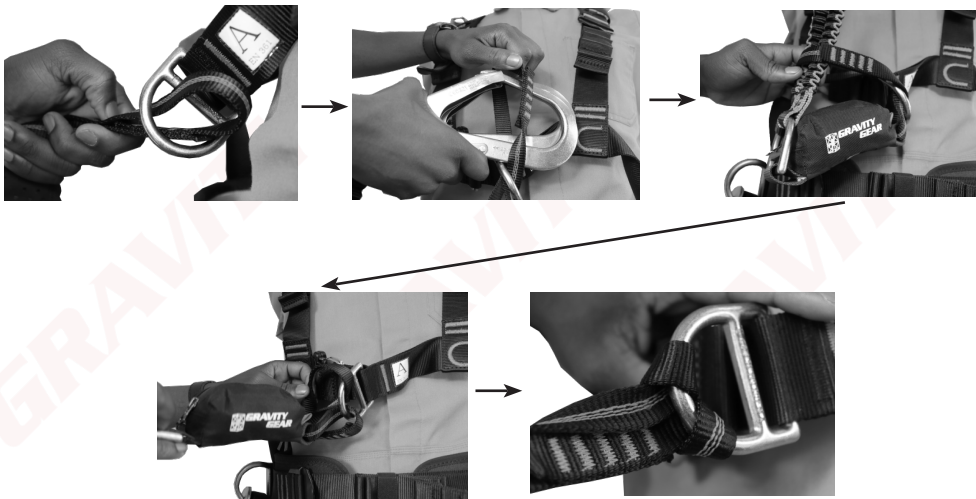


UNSAFE USE OF SHOCK
ABSORBING LANYARDS



SAFE USE OF SHOCK
ABSORBING LANYARDS

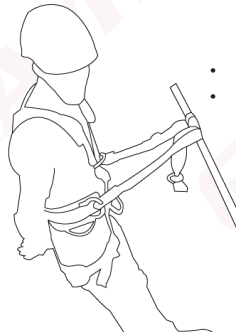
How to fit the Gravity Shock Absorbing Lanyard



Work Positioning Lanyard (EN 358)

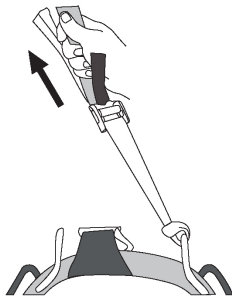
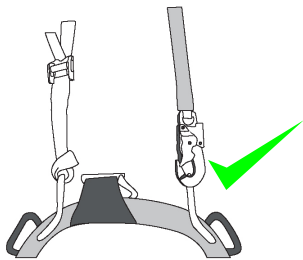
A work positioning lanyard is used to allow a worker to use both hands freely in order to do work effectively.

- MUST NEVER BE USED WITHOUT A FALL ARREST SYSTEM.
- Should be wrapped around the structure to prevent slipping.
- S.W.L. = 140 kg
- M.B.S. = 1400 kg

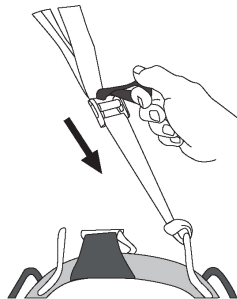
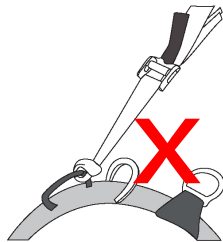


The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

How to fit the Gravity Work Positioning Lanyard



How to adjust the Gravity Work Positioning Lanyard



Rope Grabs Devices (EN 353-2)



Used on a vertical lifeline as a Fall Arrest device. This will grab the rope in the event of a fall, limiting the falling distance to only rope stretch.

- Must be used in a safe fall factor.
- Used on a vertical lifeline.
- Must not be manipulated directly as this may interfere with the working action of the device.
- S.W.L. = 120 kg
- M.B.S. = 1200 kg



Descending Devices (EN 341)



A variable friction device that controls the rate of descent. Used for an emergency escape, horizontal lifelines and rescue.

- Must be locked when not in use.
- Used during emergency escape, when tensioning a horizontal lifeline and during rescue.
- When using in rescue, extra friction must be applied.
- S.W.L. = 120 kg
- M.B.S. = 1200 kg
- S.W.L (with extra friction) = 220 kg



Slings (EN 566 & 795)

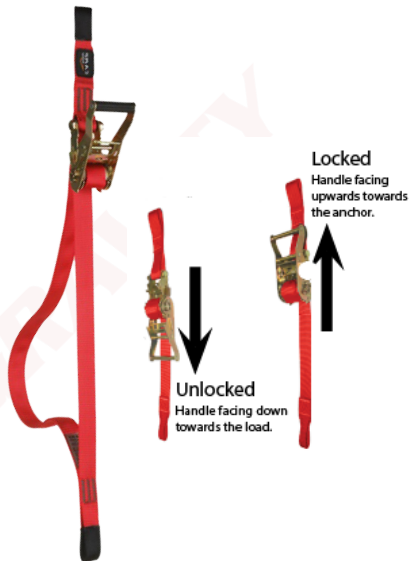


Used to anchor systems to structures when sharp edges are not present.

- Should be wrapped around the structure to prevent slipping.
- Stitching must be straight to prevent the weakening of the sling.
- Slings with PVC sleeves should be used should sharp edges be a concern.
- S.W.L. = 220 kg
- M.B.S. = 2200 kg



Rescue Ratchet Slings (EN 354)

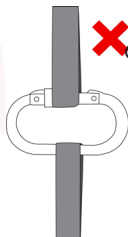
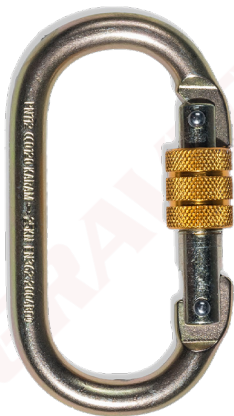


Used during the ratchet rescue to lift casualty up to allow rescuer to remove the casualty's lanyards.

- Has a ratchet mechanism to allow for easy lifting.
- Must be locked (closed completely) when not in use.
- Always reset the ratchet after use.
- S.W.L. = 150 kg
- M.B.S. = 1500 kg

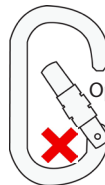
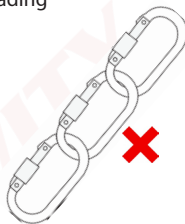
Connectors (EN 362)

Used to connect a range of devices and different pieces of equipment.



✗ Cross-loading

Chain-linking



Open gate

Gate facing downward



Opposite and opposed



- S.W.L. = 220 kg
- M.B.S. = 2200 kg

Work Ropes

Semi-Static Ropes (EN 1891)

Ribbon stating date of manufacture

Kern\Core

Sheath\Mantle



Used throughout work at height systems and is a crucial part of height safety.

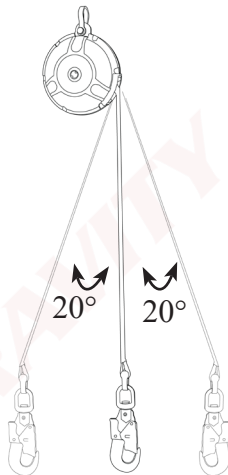
- Stretches between 5% and 10%.
- Is of kermantle construction. Consists of two parts:
Kern (Core) = 80% of the strength.
Mantle (sheath) = 20% of the strength.
- S.W.L. = 250kg
- M.B.S. = 2500kg

Retractable Lanyard (EN 360)

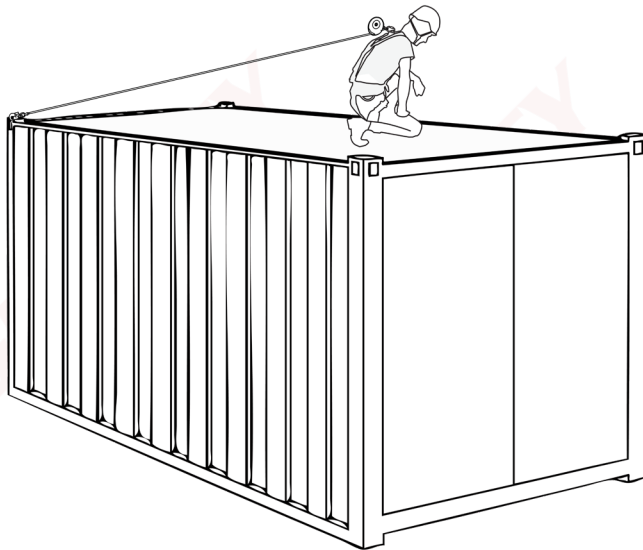


Used as a self-retracting fall arrest device, this allows quick and easy movement up and down. The device limits falling distance to approximately 30 cm.

- Similar functionality of a car safety belt; it allows a technician to freely move up and down, and locks only when a person has fallen.
- Can be used for work positioning and fall arrest.
- Some models include a shock-absorbing function.
- Has a 20 degree maximum range during vertical use.
- Never use above a Fall Factor 1.
- S.W.L. = 120 kg
- Please note the S.W.L might differ from manufacturer or by design.



Using Retractable Lanyard while working on containers



- When using it on a rooftop or container the device should connect to the technician and not to the structure, to avoid any reduction in the stopping distance. However not all Retractable are rated for this work.

Carrying Tools at Height

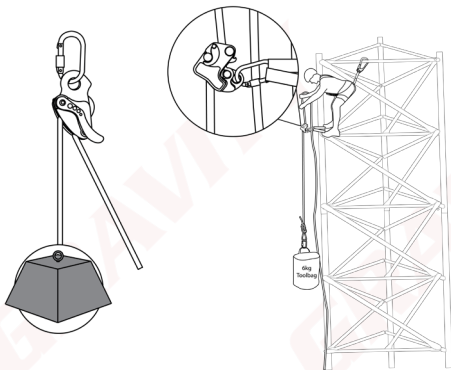
All tools must be tied off with tool lanyards or accessory cord to prevent tools from being dropped from height. Tools being dropped can cause damage to property and even injury or death to persons below.



Tools weighing 8 kg or less should be connected to the harness gear loops. Tools weighing in excess of 8 kg should be lifted using a separate 1:1 pulley system with a maximum lifting capacity of 20 kg. Any tools or equipment weighing more than 20 kg needs to be lifted by a qualified rope rigger. A qualified rope rigger may lift up to 40 kg with using a 1:1 system and up to 100 kg using a 3:1 system.

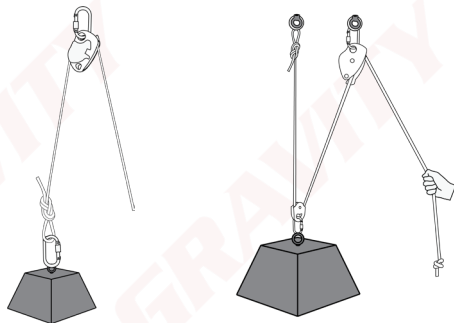
Direct Lifting

Light materials weighing 8 kg or less should be connected to the harness gear loops. Light materials weighing in excess of 8 kg should be lifted using a separate 1:1 or 2:1 pulley system with a maximum lifting capacity of 35 kg. Any light materials or equipment weighing more than 35 kg need to be lifted by a qualified rope rigger.



Principles of Slings:

- Never exceed the safe working load of equipment.
- Always protect slings from sharp edges.
- Always perform a pre-lift before lifting.
- Always consider the center of gravity.



Principles of Rigging:

- Never stand underneath a load.
- Always ensure that the route traveled is clear of obstructions.
- Ensure that the correct equipment is used for the lift.

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software



Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Hardware



Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

Equipment Storage

Equipment must be stored correctly and according to manufacturer's specifications to prevent damage to equipment when not in use.

Store Equipment:

- In a cool, dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.), which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges; use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.



Rope Bagging:

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over the structure or ladder and pull the rope into the bag, this will prevent knotting and allows a rope to be bagged faster.



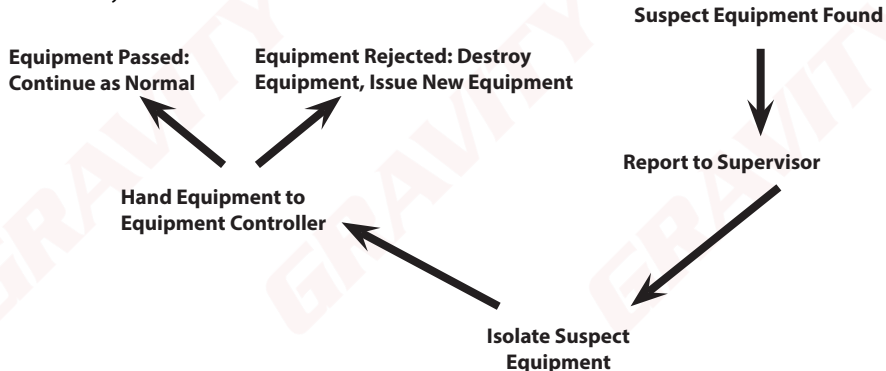
Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use, store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Reporting Suspect Equipment

Gear that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or old equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer.
NO CONTRACTOR MAY REPAIR PPE.

D. Work at Height Principles

1. Fall Factors

Fall Factor =

$$\frac{\text{Distance of fall}}{\text{Length of lanyard}}$$



Fall Factor 0:



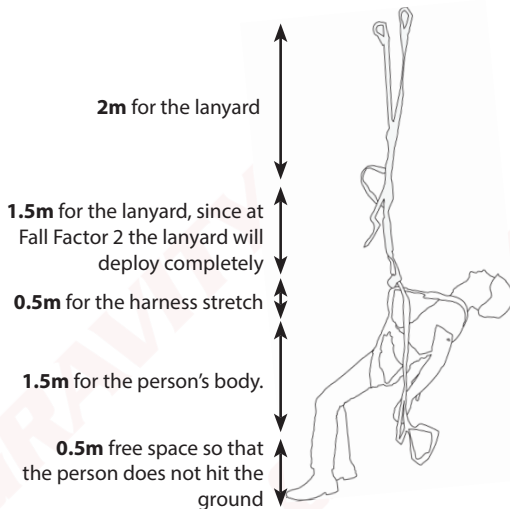
Fall Factor 1:



Fall Factor 2:

1. Minimum Free Space

Minimum free space (MFS) is the open space a technician would need below him to safely arrest a person's fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.



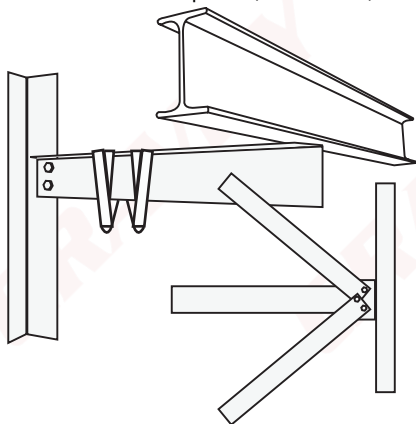
$$\text{THUS: } 2 \text{ m} + 1.5 \text{ m} + 0.5 \text{ m} + 1.5 \text{ m} + 0.5 \text{ m} = 6 \text{ m}$$

E. Anchor Points

Anchor points are intended to be the last remaining connection point between a worker and a structure should a fall occur to prevent the worker from falling to the ground.

Anchor points are mainly divided into two categories:

- Self-identified anchor points, and
- Planned anchor points (EN 795:2012).



Self-identified

- S.W.L. = 150 kg
- M.B.S. = 1500 kg



Gravity Access Anchor

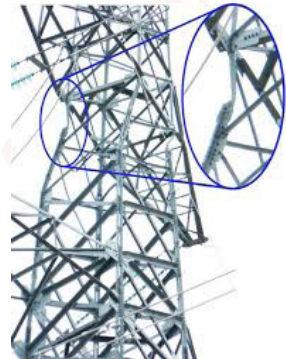
Eye-nut

Planned

- S.W.L. = 120 kg
- M.B.S. = 1200 kg

3. Structural Evaluations

- "It is the responsibility of the owner of the structure to ensure the structure is safe to climb and to perform any rigging and lifting activities when needed.
- All structures must be visually inspected before any climbing or lifting operations are conducted. It is not the responsibility of the climber or the rigger to certify the structure, but must only ensure that the structure is safe to climb. Each climber must check the following.
 - That the structure is signed off by an engineer.
 - That the fixed fall arrest system is certified and safe to use.
 - That there is no missing members or loose telecoms equipment.
 - Any roof structure must also be inspected that the surface areas are secure and safe to access."



4. Temporary Vertical Lifelines

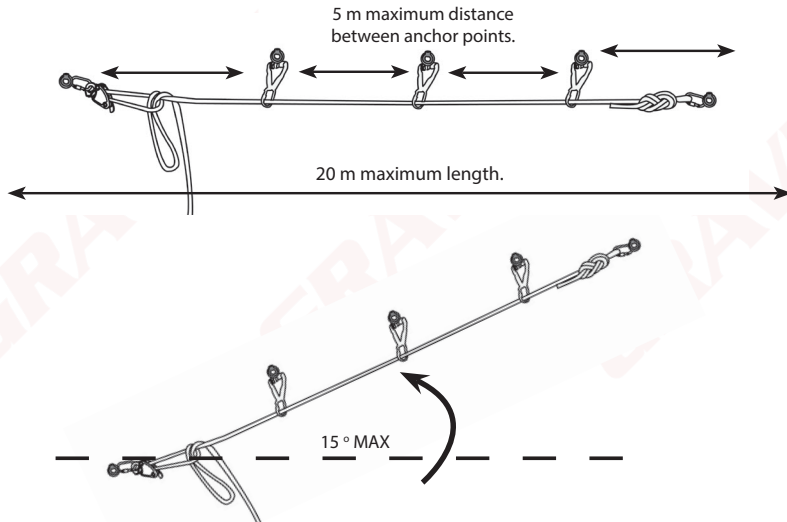
Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a rope grab device which results in a reduced falling distance, only being limited to the amount of rope stretch.

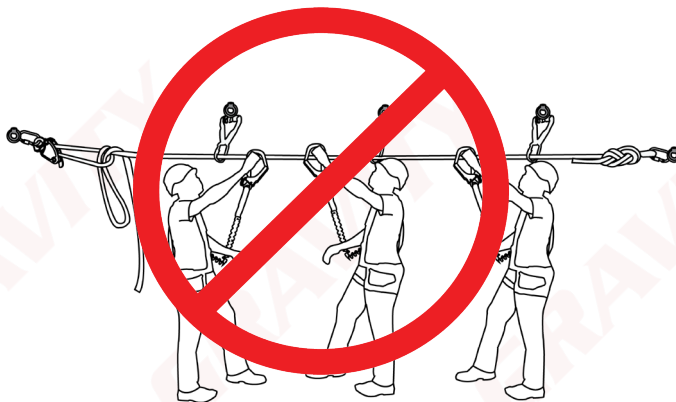
- Maximum Length: 30 m
- Maximum persons allowed on the lifeline: 1 Person.
- Maximum angle between the 2 loops is 90°.



5. Temporary Horizontal Lifelines

Horizontal lifelines are used for safe and quick movement from side to side. These lifelines are to be used with a shock absorber which can slide from side to side as the technician moves.





- No more than 2 users on a line
- No more than one user per section

F. Emergency Procedures

1. Rescues

Ensure that a rescue plan is in place to ensure that a rescue can be performed as quickly and efficiently as possible.

Perform a rescue when;

- A person has fallen and his shock-absorbing lanyard has deployed.
- A person has lost consciousness.
- A person is unable to climb down to safety unassisted.

All rescues are different but there are a few things that stay the same throughout:

- Always call the emergency services before attempting a rescue.
- Always establish and maintain communication with the casualty throughout the rescue with either: two-way radios, hand signals or simply talking if the person is within earshot.
- It is recommended that all rescuers are trained first aiders as well.

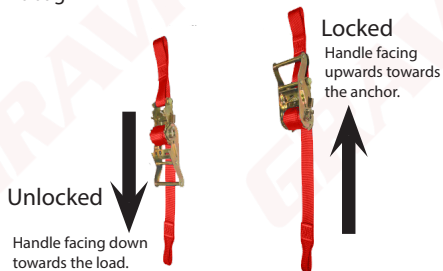
Ratchet rescue procedure:

1. Call the emergency medical services.
2. Climb up to the casualty.
3. Connect the rope to the casualty's top D-link.
4. Connect the suspension trauma sling.
5. Climb above the casualty.
6. Anchor the rescue ratchet with an attachment sling.
7. Remove all the slack between the casualty and the descending device.
8. Lift the casualty with the rescue ratchet.
9. Remove the casualty's lanyard.
10. Apply extra friction.
11. Lower the casualty.

Ratchet Rescue

Rescue Kit:

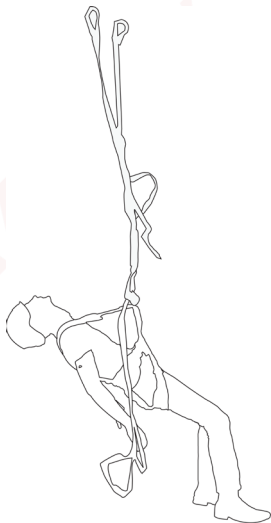
1. Attachment sling
2. Rescue ratchet
3. Descending device
4. Kernmantle rope
5. Suspension trauma sling
6. Attachment karabiner
7. Extra-friction karabiner
8. Kit bag



Requirments:

- A self-escape rescue kit to facilitate self-evacuation during an emergency must be taken up by the first person who ascends the tower, and the kit must be stored in a safe place on the crow's nest of the monopole.

2. Suspension Trauma



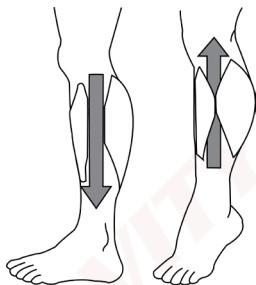
The two biggest risks associated with awaiting a rescue while suspended in a harness, are injury or death due to suspension trauma.

Due to the force of gravity, the blood in the fallen person's body is pulled toward the ground, and pools in the legs.

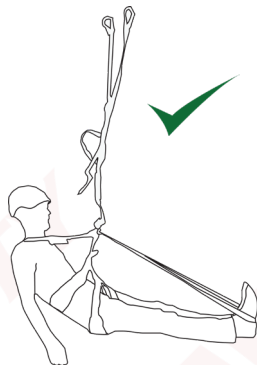
Due to the constriction of the leg loops of the harness, the blood cannot return to the fallen person's heart. This prevents blood from being circulated through a person's body. Thus, less and less blood reaches the brain, leading to loss of consciousness and can lead to brain damage. Finally, this can lead to cardiac arrest and eventually death.

All of this can happen within 15-20 minutes; so it is important to have a good knowledge of the techniques needed to prevent suspension trauma, for yourself, and the casualty that needs to be rescued.

Preventing Suspension Trauma



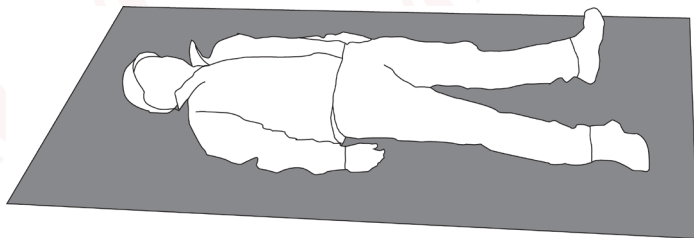
Muscle Pump: Moving your legs, feet and toes will contract the muscles in the legs and force blood back into circulation, which provides the brain with the much-needed oxygen to stay conscious.



Suspension Trauma Sling: Will allow you to stand up or to get horizontal, prevent the blood pooling in the legs as well as relieve the pressure of the leg loops which will allow the blood to remain in circulation to the heart and brain. Should the casualty be unconscious, the rescuer can do this for the casualty.

3. Supine Recovery Position

This position allows the blood to flow slowly enough as to prevent Reflow Syndrome, while at the same time keeping the casualty stable and still as to prevent any possible further injury to the casualty's neck and/or spine. This position is also known as the "Supine Position".



G. Knots

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot will weaken the rope.

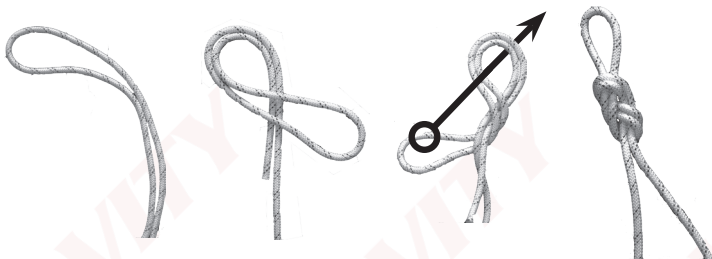
1. Figure 8 Stopper Knot



This knot is made in the end of a rope to stop people and equipment from being able to move off the end of the rope.

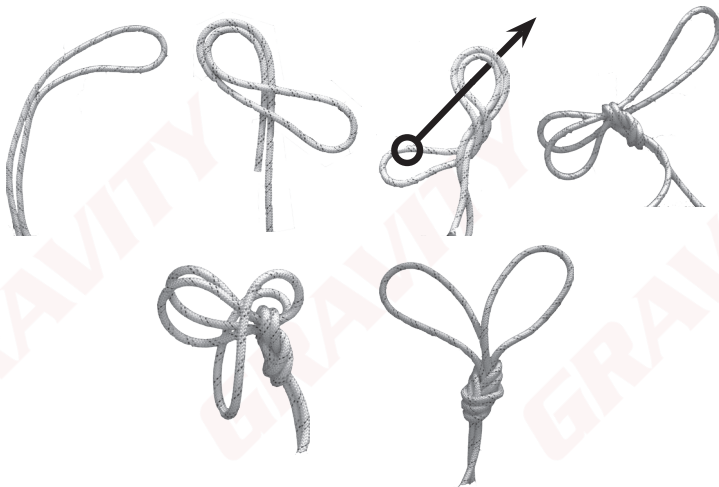
The tail must be longer than your fist, but shorter than your forearm.

2. Figure of 8 on a Bight



This knot is used to connect the rope to a single anchor point or devices and is used for the installation of a horizontal lifeline as well as during the rescue process.

3. Double Loop Figure 8



This knot is used to connect a rope to two separate anchor points, most commonly used to rig vertical lifelines.

4. Klemheist

Used to create an anchor point on a vertical or diagonal pole. Must be used in a Fall Factor 1 or better.



H. Specialised content

1. Monopole internal climbing and rescue procedure

Climbing in monopoles requires specific fall arrest and rig-for-rescue equipment that must be used. Before entering a monopole, the inside of the monopole must be inspected for bees, loose electrical and any other hazards that can pose a risk to the team's safety.

Minimum equipment for climbing and rescue from below and above:

2 x Full Body Harness	- EN361/358
2 x Shock Absorbing Lanyard (SAL)	- EN355 (Length of the shock absorber must not be longer than 1 meter)
2 x Work Positioning Lanyard	- EN358
2 x Climbing Helmet	- EN12492 (Consideration has been given to not use EN 397 Industrial helmets)
2 x 120 cm Open Round Sling	- EN566
1 x Low-stretch Kernmantle Rope	- EN1891 (Length of the rope must be double the height of the tower)
2 x Double-Locking Descending Device	- EN341
10 x Auto-lock Carabiner (ALC)	- EN362
2 x Rescue Sling	- EN566
2 x Pylon Hooks	- EN362

NB: Before climbing, ensure that all PPE is suitable for the type of work that needs to be performed.

The harness must be suitable for climbing inside a monopole, and PPE must be appropriate to perform a rescue within the monopole should the need arise.

Procedure for the first climber:

1. Climber 1 makes a figure 8 on a bight on one end of the working rope and connect it to an auto-lock carabiner (ALC), and then connects to the top d-link of his harness.
2. Climber 1 attaches one end of the rescue sling on his back connection point and the other end of the sling on the working rope.
3. Climber 1 installs the shock absorber lanyard (SAL) to his side while connecting two separate pylon hooks, each with its own carabiners, to the working end of the rope. The working rope must be connected through both carabiners that are connected to the pylon hooks.
4. Climber 1 takes a 120 cm sling, two carabiners, and one descender by connecting it to the side d-link.
5. Climber 2 installs a descending device on a suitable anchor point (around 2 rungs) with the use of a 120 cm sling and auto-lock carabiner (see Figure 1) at to the right.
6. Climber 2 installs the rope in the descending device.
7. Climber 1 to ascend the structure with the use of the two pylon hooks attached to the working rope (see Figure 1).
8. Climber 2 feeds the rope through the descender to allow for rope slack of not more than 1 meter for climber 1 to move the pylon hooks.
9. In the event of a fall, climber 2, who is at the bottom controlling the descender, can lower climber 1 down to safety (as per Figure 2.3 and 2.2)
10. Once climber 1 has ascended the monopole safely, he can connect the shock absorber on his side to a secure anchor point and move outside to the crow's nest. This should be done before removing the work rope with the auto-locking carabiner from the chest d-link.

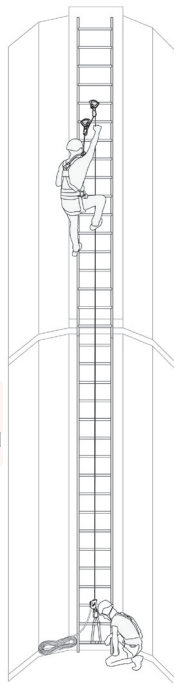


Figure 1

11. When climbing down, climber 1 connects himself back onto the system and climbs down while climber 2 belays him down. Follow steps 1–10 for the first climber before starting with the below steps.

Figure 2

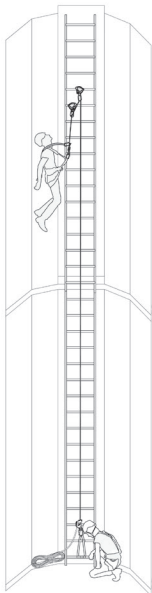


Figure 2.1



Figure 2.2



Figure 2.3

Procedure for the second climber

- Follow steps 1–10 for the first climber before starting with the below steps.
12. Climber 1 sets up a belay system for climber 2 by securing the sling at the top (around two rungs) of the structure, connecting the secondary descending device to the sling and placing the end of the rope inside the descending device (see Figure 3 to the right).
 13. Climber 2 at the bottom then disconnects the descender, sling and carabiner, takes the end of the rope and makes a figure 8 on a bight knot and connects it to an auto-locking carabiner and then to the chest d-link.
 14. Climber 2 then connects the rescue sling that is connected to his back connection point on his harness and secures the other end of the rescue sling to the rope then climber 1 belays climber 2 up to safety as per Figure 3.
 15. While climber 2 ascends, climber 1 pulls the rope through the descender to remove the slack in the rope that goes to climber 2.
 16. Should climber 2 fall, the distance will be minimal due to the small amount of slack between the climber and the descender (see Figure 4).
 17. Climber 2 can then also be lowered to safety by climber 1.
 18. If climber 2 does not fall and is able to climb all the way to the top, no rescue will be needed.
 19. When descending the tower, climber 1 can assemble the same belay system as per Figure 5 and belay climber 2 down the monopole (see Figure 5).

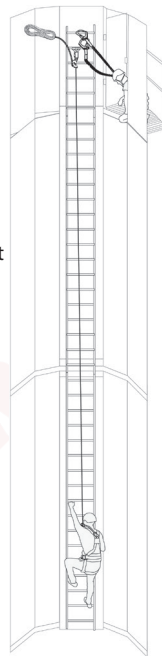


Figure 3

20. Climber 2 then assembles the belay system at the bottom of the tower and belays climber 1 down.
21. Climber 1, once again, makes a figure of 8 on a bight knot and connects it to the auto-locking carabiner, and then connects it to his chest d-link.
22. Climber 1 connects the pylon hooks with an auto-lock carabiner and secures it to the work rope, then remove the shock absorber that was secure and climbs down to safety.

Figure 4

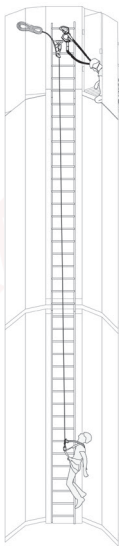


Figure 5

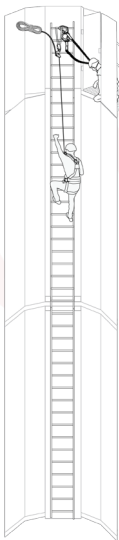


Figure 6



2. Emergency self-escape rescue procedure

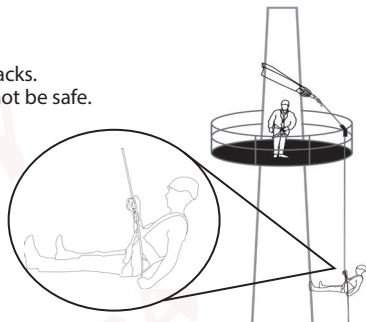
During work at height, emergency scenarios may develop where a worker needs to self-evacuate immediately.

Scenarios may include, but are not limited to:

- Fire or smoke on the inside of a monopole or sectional pole.
- Obstruction inside a monopole.
- Sudden onset of external conditions, e.g. veld fires or bee attacks.
- Inclement weather, where climbing down the ladder would not be safe.
- Birds of prey on the crow's nest.

1. Secure the anchor sling around the spine of the monopole above the crow's nest.
2. Make a figure 8 on a bight knot on one end of the rope and secure it with a carabiner to the anchor sling.
3. Makes a figure 8 on a bight knot on the other end of the rope that will be lowered to the ground.
4. Secure the edge protection around the rope and move it to any surface area that has a sharp or hot surface.
5. Place the rope into the descending device and secure the descending device to his/her chest d-link with a carabiner.
6. Do a quick function test to ensure that the descending device is working correctly before removing the SAL.
7. Technician can no descend on the rope by placing one hand on the lever of the descender and the other hand on the rope to allow a safe descent.
8. Once safely on the ground, detach from the descending device by disconnecting the carabiner from the chest d-link.
9. Remove the descending device from the rope, connect the carabiner to the descender and secure it on the figure 8 on a bight that is on the bottom end of the rope.
10. The next technician then pulls the rope up and follows the same procedure from step 5 onwards.

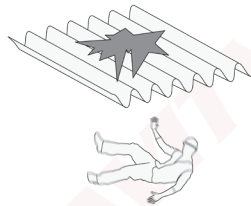
NB: Ensure that proper communication is maintained with all people above and below.



3. Roof work

Roof work entails the deed of installing, removing, repairing or otherwise maintaining any covering above ground level whether on a roof or in a roof (for example, building roofs with tiles or zinc plates, water proofing, painting, installing and maintaining solar systems, plumbing, gutter and chimney cleaning etc.).

Working on roofs is a dangerous activity and many fatalities occur due to workers not being aware of the risks and not mitigating the relevant risks properly.



Main hazards associated with roof work:

- Falling from a flat or sloping roof due to inadequate edge protection or barricading.
- Falling through fragile/damaged roof surfaces, for example skylights, ceilings etc.
- Falling objects from a roof due to inadequate edge protection or objects not being tied-off.
- Falling from a ladder or alternative access method when accessing a roof.
- Manual handling of materials caused by the size, weight, temperature or chemical natures of materials.

Every roof work site will have its challenges that need to be assessed before work may start. For example, the material of the roof, the access method to the roof, and the system to ensure the worker's safety all play a huge role and will have to be assessed before work starts.

Should there be any question regarding the integrity of the structure holding up the roof (trusses, beams etc) or the roof covering (iron sheeting, tiles etc), work should be halted until a structural engineer has confirmed that the roof is safe to work on.

3.1 Access methods for roof work:

Portable ladders

Portable ladders are the most common and cheapest way to gain access to roofs. However, it is also considered among the most dangerous methods of roof access if the worker is not trained properly. A portable ladder use training should be attend to obtain competence.



Scaffolding

Scaffolding is one of the most expensive and most time-consuming roof work option. Scaffolding must be assembled by a competent erector and signed off by a competent inspector.



Mobile elevated work platforms (MEWPs)

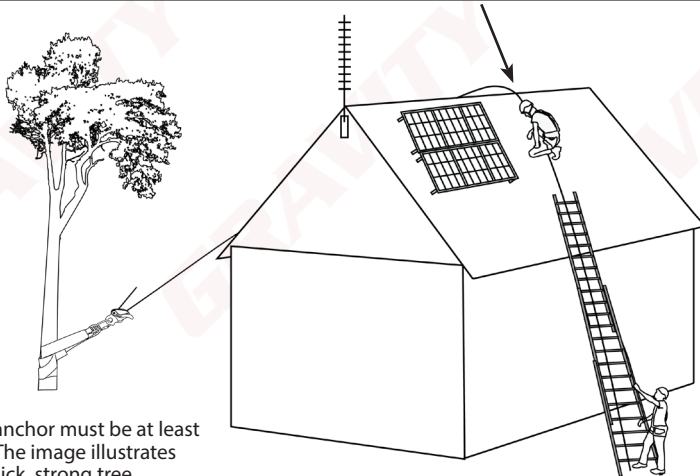
MEWPs are probably the safest method, however, it is also expensive and the site does not always allow for a MEWP to gain access or to move around. A competent MEWP operator needs to operate the MEWP at all times.



3.2 Safety systems while working on roofs with anchorage methods:

NB: The anchorage methods used in the system below have to meet the requirements of EN795 type B or E. It also allows the user to only be on one side of the roof and will not be allowed to go with in 0.5m of the knock.

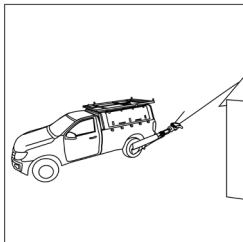
Please note that the following methods can only be used for a pitch no more than 30 degree, any slopes steeper than 30 degrees require rope access techniques.



A self-identified anchor must be at least 1,200 kg strong. The image illustrates anchoring to a thick, strong tree.

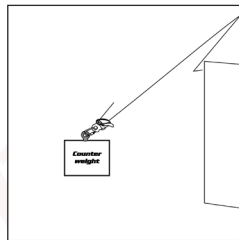
3.3 Various anchorage options

Anchoring to a vehicle.



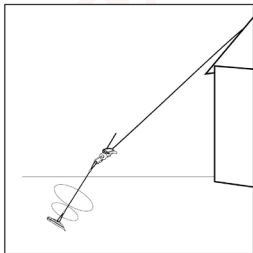
This system is installed on the vehicle axle. The climber should be in possession of the vehicle keys.

Anchoring to a weight designed to carry the load of a self-identified anchor point.



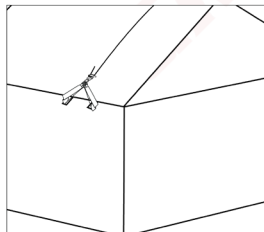
This system is installed on a weight designed to carry the load of self-identified anchor point.

Anchoring to a duckbill anchor.



This system is installed into the ground and then pulled to engage with a wire sling attached.

Anchoring to two trusses of the roof.

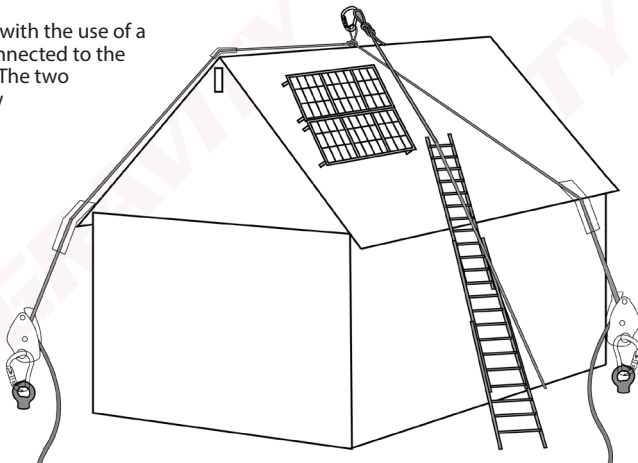


This system is installed on two of the trusses to share the weight between the anchor points.

3.4 Low level FAS system.

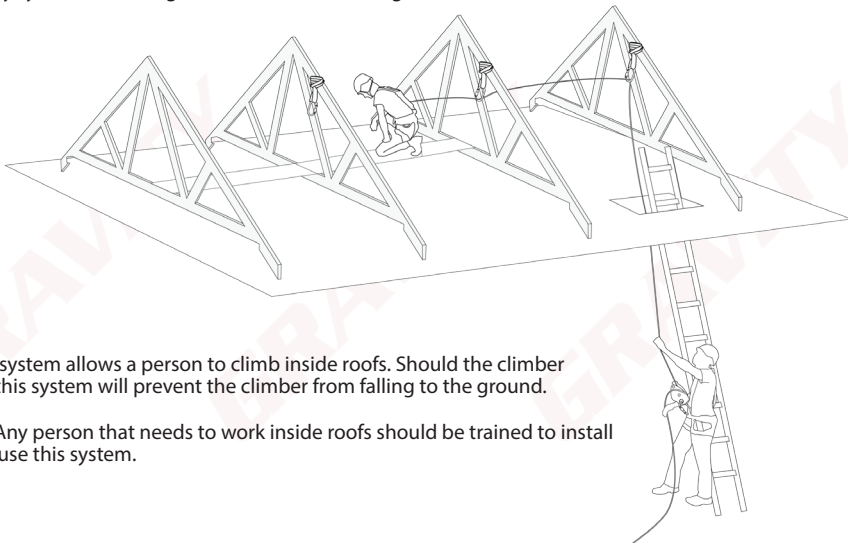
NB: The low level FAS system below has to meet the requirements of EN795 type B or E. It also allows the user to move to any side of the roof.

The user will connect themselves with the use of a rope grab to the lifeline that is connected to the knot in the middle of the system. The two descenders attached to the below anchors enables the tensioning of the system, as well as a rescue at any stage.



3.5 Working inside roofs.

Safety systems anchorage methods while working in roof:



This system allows a person to climb inside roofs. Should the climber fall, this system will prevent the climber from falling to the ground.

NB: Any person that needs to work inside roofs should be trained to install and use this system.

The Way Forward

Congratulations on completing Gravity Training's Fall Arrest and Basic Rescue course.

At Gravity we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for one or more of the below courses:

US 14706 - Rope Rigging

US 229994 - Fall Protection Plan Development

US 229998 & 230000 - Rope Access Level 1

Portable Ladder User

Climbing Equipment Inspection and Management